HACKFLASH HACKATHON

Problem Statement 1

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1. EXECUTIVE SUMMARY

This project analyses hospital admission data to understand patient demographics, common medical conditions, admission patterns, billing amounts, and length of hospital stays. Exploratory Data Analysis (EDA) was performed using Python and its data analysis and visualisation libraries.

The original dataset contained 55,500 records and 15 columns. During preprocessing, 534 duplicate rows were identified and removed, leaving 54,966 unique rows. Missing values, invalid dates, unusual billing amounts, age ranges, and text formatting were also examined.

The analysis found that arthritis was the most frequently recorded medical condition, although the six medical conditions had relatively similar numbers of records. Emergency admissions had the longest average hospital stay, at approximately 15.58 days. The correlation between billing amount and length of stay was approximately -0.0049, indicating almost no linear relationship in this dataset.

The findings provide an overview of the dataset and highlight patterns that could be investigated further by hospital administrators. However, the results describe associations in the available data and should not be interpreted as proof of cause and effect.

2. PROBLEM UNDERSTANDING

The objective of this project is to explore hospital admission records and identify patterns that may help healthcare organisations better understand their patient population, medical conditions, admissions, and billing.

The project addresses the following questions:

1. What are the demographic characteristics of the patients?
2. Which medical conditions occur most frequently?
3. How are admissions distributed across admission types?
4. Do billing amounts differ across medical conditions?
5. Do hospitals show differences in average billing and length of stay?
6. Do insurance providers show differences in average billing?
7. What patterns appear in medication and test result records?
8. Is there a relationship between the length of hospital stay and billing amount?
9. Are there data quality issues that need to be considered before analysis?

The analysis uses descriptive statistics, frequency counts, grouped summaries, correlation analysis, and visualisations to answer these questions.

3. DATASET OVERVIEW

The dataset contains hospital admission records with information about patients, medical conditions, admission and discharge dates, hospitals, insurance providers, medications, test results, and billing amounts.

The original dataset contains 55,500 rows and 15 columns.

The columns are:

- Name
- Age
- Gender
- Blood Type
- Medical Condition
- Date of Admission
- Doctor
- Hospital
- Insurance Provider
- Billing Amount
- Room Number
- Admission Type
- Discharge Date
- Medication
- Test Results

A derived column named "Length of Stay" was created by calculating the difference between the discharge date and the admission date. This increased the dataset to 16 columns.

The dataset contains a mixture of numerical, categorical, and date columns.

Numerical columns include Age, Billing Amount, Room Number, and Length of Stay.

Categorical columns include Gender, Blood Type, Medical Condition, Hospital, Insurance Provider, Admission Type, Medication, and Test Results.

Date columns include Date of Admission and Discharge Date.

4. DATA PREPROCESSING AND QUALITY AUDIT

Data preprocessing was performed before the exploratory analysis to improve data quality and ensure that the analysis was reproducible.

4.1 Duplicate Records

The original dataset contained 534 duplicate rows. These records were removed using Pandas drop_duplicates().

The dataset decreased from 55,500 rows to 54,966 rows.

After removing duplicates, a second duplicate check confirmed that no exact duplicate rows remained.

Reason: Duplicate records can affect frequency counts, summary statistics, and visualisations by giving repeated records additional weight.

4.2 Missing Values

The missing-value audit identified two missing values in the Billing Amount column. All other original columns and the derived Length of Stay column had no missing values.

The two missing billing records were retained because removing the complete records would also discard valid information about the patients and their admissions.

Billing calculations automatically exclude missing billing values when calculating statistics such as the mean.

Reason: Retaining these records preserves useful information while allowing billing analysis to use the available values.

4.3 Date Conversion

The Date of Admission and Discharge Date columns were converted into datetime format using Pandas.

This conversion allows date subtraction and chronological validation.

No missing dates were found after conversion.

4.4 Length of Stay

A new column named Length of Stay was calculated by subtracting Date of Admission from Discharge Date.

The formula is:

Length of Stay = Discharge Date - Date of Admission

The resulting number represents the difference in days between the two dates.

The length-of-stay summary was:

- Number of records: 54,966
- Mean: 15.50 days
- Median: 15 days
- Minimum: 1 day
- Maximum: 30 days
- Negative stays: 0

- Zero-day stays: 0
- Missing stays: 0

No discharge-before-admission records were identified.

4.5 Age Validation

Patient ages ranged from 13 to 89 years.

The mean age was approximately 51.54 years, and the median age was 52 years.

No ages below zero or above 120 were found.

Therefore, no age records were removed.

4.6 Negative Billing Amounts

The audit identified 106 negative billing amounts, representing approximately 0.19% of records with available billing values.

The negative values ranged from approximately -2,008.49 to -23.87. Their average was approximately -502.47.

Negative billing values may represent data-entry problems, adjustments, refunds, or other accounting situations. However, their actual meaning cannot be confirmed from the available columns.

These records were retained rather than automatically changed or deleted.

Reason: Removing or replacing negative values without understanding their meaning could introduce bias into the analysis. Further validation against the original billing system would be required.

4.7 Text Formatting and Patient Names

A text-formatting audit found 5,978 names with possible casing variations. For example, the same-looking name could appear with different uppercase and lowercase letters.

There were 49,992 distinct original name strings and 40,235 distinct names after normalising case and surrounding spaces.

The difference of 9,757 does not establish that 9,757 duplicate patients exist. Names alone are insufficient to confirm patient identity.

Original names were preserved, and name normalisation was used for auditing purposes rather than as proof that records belonged to the same person.

The hospital column contained 39,876 distinct hospital-name strings. This unusually high number may indicate inconsistent naming or synthetic data generation. Similar-looking hospital names were not merged because there was no verified mapping to confirm that they referred to the same hospital.

4.8 Final Preprocessing Result

After duplicate removal and creation of the Length of Stay column, the final dataset contained 54,966 rows and 16 columns.

The two missing billing values and 106 negative billing amounts were documented and retained.

No invalid ages, negative stays, zero-day stays, or discharge-before-admission records were identified.

5. EXPLORATORY DATA ANALYSIS

5.1 Patient Demographics

Age:

- Mean age: 51.54 years
- Median age: 52 years
- Minimum age: 13 years
- Maximum age: 89 years

The mean and median ages are close, suggesting that the central age distribution is similar under these two measures. The dataset includes patients across a wide age range.

Gender:

- Male: 27,496 records, approximately 50.02%
- Female: 27,470 records, approximately 49.98%

The gender distribution is almost evenly balanced.

Blood Type:

The most frequently recorded blood type was A-, with 6,898 records. The least frequent was O-, with 6,804 records.

The differences between blood-type counts were relatively small, indicating a fairly even distribution in the dataset.

5.2 Distribution of Medical Conditions

The six medical conditions were compared using their record counts.

- Arthritis: 9,218 records
- Diabetes: 9,216 records
- Hypertension: 9,151 records
- Obesity: 9,146 records
- Cancer: 9,140 records
- Asthma: 9,095 records

Arthritis was the most frequently recorded condition, while asthma had the lowest count.

The difference between the highest and lowest counts was only 123 records. Therefore, no single condition overwhelmingly dominated the dataset.

These figures describe the distribution of conditions in the available records. They should not be treated as estimates of disease prevalence in the general population.

5.3 Billing Amount by Medical Condition

Average billing amounts were calculated for each medical condition.

- Obesity: approximately 25,804.36
- Diabetes: approximately 25,660.48
- Asthma: approximately 25,633.46
- Arthritis: approximately 25,513.63
- Hypertension: approximately 25,501.52
- Cancer: approximately 25,152.32

Obesity had the highest average billing amount, while cancer had the lowest.

The difference between the two averages was approximately 652.04.

The average billing amounts were relatively close compared with the overall variation in individual billing amounts. These results do not establish that a particular medical condition causes higher or lower bills.

Other factors, such as individual treatment, services received, insurance arrangements, and patient characteristics, may also influence billing.

5.4 Admission Type Analysis

The admission records were grouped into three types.

- Elective: 18,473 records, approximately 33.61%
- Urgent: 18,391 records, approximately 33.46%
- Emergency: 18,102 records, approximately 32.93%

The three categories had similar numbers of admissions.

Average length of stay by admission type:

- Emergency: 15.58 days
- Elective: 15.51 days
- Urgent: 15.40 days

Emergency admissions had the longest average stay, while urgent admissions had the shortest.

The difference between the emergency and urgent averages was approximately 0.18 days. This is a small difference and should not be interpreted as statistically significant without further testing.

Average billing by admission type:

- Elective: approximately 25,613.07
- Urgent: approximately 25,513.77
- Emergency: approximately 25,505.33

Elective admissions had the highest average billing, while emergency admissions had the lowest. The difference was approximately 107.74.

The relatively small differences suggest that the average billing amounts were similar across admission types in this dataset.

5.5 Hospital-Level Analysis

Hospital records were grouped by hospital name to compare patient counts, average billing, median billing, and average length of stay.

Because many hospital names had very few records, only hospital-name groups with at least 20 records were included in the main comparisons.

This threshold helps reduce the influence of very small groups, although it does not resolve inconsistent hospital naming.

Among the included groups, Johnson Ltd had the highest average billing, approximately 34,735.49, based on 21 records.

Smith Group had the lowest average billing, approximately 22,406.42, based on 36 records.

The highest average length of stay was observed for Brown PLC, at approximately 21.14 days, based on 22 records.

These comparisons show that differences exist between the hospital-name groups. However, they do not prove that one hospital provides better or worse services or is more expensive because patient characteristics, treatment requirements, and data quality may differ.

Hospital-name standardisation and additional information about patient case mix would be required for a reliable hospital performance comparison.

5.6 Billing Amount by Insurance Provider

Average billing amounts were compared across insurance providers.

- Medicare: approximately 25,628.32
- Blue Cross: approximately 25,603.46
- Aetna: approximately 25,551.27
- Cigna: approximately 25,526.00
- UnitedHealthcare: approximately 25,413.22

Medicare had the highest average billing amount, while UnitedHealthcare had the lowest.

The difference was approximately 215.10.

The average amounts were relatively similar across providers. These figures represent recorded billing amounts and should not be interpreted as insurance premiums, amounts paid by insurers, or patients' out-of-pocket expenses.

Further information would be required to determine how insurance arrangements relate to actual payments.

5.7 Medication Analysis

The medication counts were:

- Lipitor: 11,038 records
- Ibuprofen: 11,023 records
- Aspirin: 10,984 records
- Paracetamol: 10,965 records
- Penicillin: 10,956 records

Lipitor had the highest count, while Penicillin had the lowest. The difference was 82 records.

The counts were relatively similar, so no single medication dominated the dataset.

These counts describe recorded medication categories. They do not indicate medication effectiveness, suitability, or clinical outcomes.

5.8 Test Results by Medical Condition

The Test Results column contained three categories: Abnormal, Normal, and Inconclusive.

Overall counts:

- Abnormal: 18,437
- Normal: 18,331
- Inconclusive: 18,198

The percentages of test results were also compared across medical conditions.

Arthritis had the highest abnormal-result percentage, at approximately 34.24%.

Hypertension had the lowest abnormal-result percentage, at approximately 32.53%.

Asthma had the highest normal-result percentage, at approximately 34.26%.

Hypertension had the highest inconclusive-result percentage, at approximately 33.53%.

The percentages were relatively close across conditions. Therefore, the descriptive analysis does not show a large difference in test-result distribution between the conditions.

Further statistical testing would be needed to determine whether the observed differences are meaningful.

5.9 Relationship Between Length of Stay and Billing

The relationship between length of stay and billing amount was investigated using correlation analysis and visualisation.

The Pearson correlation coefficient was approximately -0.004857.

A correlation close to zero indicates almost no linear relationship between the two variables in this dataset.

Average billing also varied across different lengths of stay. For example:

- A 14-day stay had an average billing amount of approximately 26,247.55.
- An 18-day stay had an average billing amount of approximately 24,835.03.

The average billing did not consistently increase as length of stay increased.

Therefore, the available data does not show a meaningful linear relationship between the number of days spent in hospital and billing amount.

However, correlation alone cannot establish that length of stay has no effect on cost. A non-linear relationship or relationships involving other variables may still exist.

6.VISUALISATIONS

The following visualisations were used or prepared in the notebook to support the analysis.

1. Age Distribution Histogram

Purpose: To understand the distribution of patient ages and identify the most common age ranges.

2. Gender Count Plot

Purpose: To compare the number of male and female patient records.

3. Medical Condition Count Plot

Purpose: To identify the most and least frequently recorded medical conditions.

4. Billing Amount Box Plot

Purpose: To show the distribution and spread of billing amounts and identify unusual values.

5. Admission Type Count Plot

Purpose: To compare the number of elective, urgent, and emergency admissions.

6. Average Billing by Medical Condition

Purpose: To compare mean billing amounts across the six medical conditions.

7. Average Length of Stay by Admission Type

Purpose: To compare average hospital stays across admission categories.

8. Hospital Billing Comparison

Purpose: To compare average billing amounts among hospital-name groups with at least 20 records.

9. Insurance Provider Billing Comparison

Purpose: To compare average billing amounts across insurance providers.

10. Length of Stay Versus Billing Scatter Plot

Purpose: To inspect whether billing tends to increase or decrease as length of stay increases.

11. Average Billing by Length of Stay Line Plot

Purpose: To examine changes in average billing across different stay durations.

The charts should be generated by executing the corresponding code in Google Colab. They should be reviewed before submission to ensure that the displayed charts match the written interpretations.

7. KEY FINDINGS AND RECOMMENDATIONS

7.1 Key Findings

1. The dataset contained 55,500 original records. After removing 534 duplicate rows, 54,966 records remained.
2. The patient gender distribution was almost evenly balanced, and the average patient age was approximately 51.54 years.
3. Arthritis was the most frequently recorded medical condition, but the six conditions had relatively similar record counts.
4. Average billing amounts were similar across medical conditions, admission types, and insurance providers.
5. Emergency admissions had the longest average stay, at approximately 15.58 days, but the differences between admission types were small.
6. Hospital-name groups showed differences in average billing and length of stay. However, the unusually large number of distinct hospital names limits the reliability of direct hospital comparisons.
7. The Pearson correlation between billing amount and length of stay was approximately -0.0049, showing almost no linear relationship in the dataset.
8. Data quality issues included 534 duplicate rows, two missing billing values, 106 negative billing amounts, and possible inconsistencies in names and hospital labels.

7.2 Recommendations

1. Improve data validation by checking for duplicate records, missing values, invalid dates, and unusual billing amounts when data is entered into the hospital system.
2. Investigate negative billing values with the finance team to determine whether they represent refunds, adjustments, or data-entry errors.
3. Establish consistent hospital naming conventions and maintain a verified hospital identifier to improve hospital-level reporting.
4. Use patient identifiers rather than names alone when checking for duplicate patient records, while following appropriate privacy and security requirements.
5. Investigate differences in hospital billing using additional information such as treatment type, patient severity, services received, and insurance payment details.
6. Conduct further analysis of length of stay using patient characteristics, medical conditions, and admission types to determine whether these variables help explain differences in hospital stays.
7. Apply appropriate statistical tests before claiming that differences between groups are statistically significant.

These recommendations are based on the observed patterns and data quality issues. They should be treated as areas for further investigation rather than confirmed explanations.

8. LIMITATIONS

The analysis has several limitations.

First, the dataset contains unusually many distinct hospital names, which may indicate inconsistent naming or synthetic data. This limits the reliability of hospital-level comparisons.

Second, two billing amounts are missing and 106 billing amounts are negative. The actual meaning of the negative values cannot be confirmed without additional financial information.

Third, the analysis is descriptive. It identifies patterns but does not establish cause-and-effect relationships.

Fourth, the dataset does not provide enough information to explain every difference in billing, treatment, or length of stay. Factors such as patient severity, specific procedures, and actual payments may be unavailable.

Finally, no formal statistical significance tests were performed for the group comparisons described in this report. Small differences should therefore not be assumed to be meaningful.

9. CONCLUSION

This project used Python to perform data preprocessing, data quality auditing, exploratory data analysis, and visualisation on a hospital admission dataset.

After removing duplicate rows and creating the Length of Stay column, the dataset contained 54,966 records and 16 columns. The analysis examined patient demographics, medical conditions, admission types, hospital groups, insurance providers, medications, test results, and billing amounts.

The findings showed relatively balanced demographic and admission-type distributions. Arthritis had the highest condition count, while obesity had the highest average billing amount among the medical conditions. Differences between insurance providers and admission types were relatively small.

The correlation between length of stay and billing amount was approximately -0.0049, indicating almost no linear relationship in the available records.

Overall, the project demonstrates how data cleaning, descriptive statistics, and visualisation can be used to identify patterns and data quality issues in hospital records. Further analysis with more reliable hospital identifiers and additional clinical and financial information could provide deeper insights.

10. CODES

The following code provides a reproducible pipeline for the data audit and exploratory analysis. It should be checked against the final Google Colab notebook before submission. Update the filename if your dataset has a different name.

10.1 Import Libraries and Load Dataset

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Load the hospital dataset.
df = pd.read_csv("hospital_data.csv")

# Display the dataset dimensions and first records.
print("Original dataset shape:", df.shape)
display(df.head())

# Review column names, data types and non-null counts.
df.info()
```

10.2 Initial Data Quality Audit

```
# Check the number of duplicate rows.
print("Duplicate rows:", df.duplicated().sum())

# Check missing values in every column.
print("\nMissing values:")
print(df.isnull().sum())

# Review numerical summary statistics.
display(df.describe())

# Review categorical values.
for col in df.select_dtypes(include="object").columns:
    print("\n", col)
    print(df[col].value_counts(dropna=False).head(10))
```

10.3 Remove Duplicate Rows

```
# Remove exact duplicate rows and preserve the original dataset.
df_clean = df.drop_duplicates().copy()
```

```
print("Shape after removing duplicates:", df_clean.shape)
print("Remaining duplicate rows:", df_clean.duplicated().sum())
```

10.4 Convert Dates and Calculate Length of Stay

```
# Convert admission and discharge dates into datetime values.
df_clean["Date of Admission"] = pd.to_datetime(
    df_clean["Date of Admission"],
    format="%d/%m/%Y",
    errors="coerce"
)

df_clean["Discharge Date"] = pd.to_datetime(
    df_clean["Discharge Date"],
    format="%d/%m/%Y",
    errors="coerce"
)

# Calculate the number of days between admission and discharge.
df_clean["Length of Stay"] = (
    df_clean["Discharge Date"] -
    df_clean["Date of Admission"]
).dt.days

# Review the calculated length of stay.
display(df_clean["Length of Stay"].describe())
```

10.5 Validate Ages, Dates and Billing

```
# Check for implausible ages.
invalid_ages = (
    (df_clean["Age"] < 0) |
    (df_clean["Age"] > 120)
)
```

```
print("Invalid ages:", invalid_ages.sum())

# Check for negative and zero-day stays.
print(
    "Negative stays:",
    (df_clean["Length of Stay"] < 0).sum()
)

print(
    "Zero-day stays:",
    (df_clean["Length of Stay"] == 0).sum()
)

# Check for discharge dates before admission dates.
invalid_dates = (
    df_clean["Discharge Date"] <
    df_clean["Date of Admission"]
)

print("Discharge before admission:", invalid_dates.sum())

# Inspect negative billing values without deleting them.
negative_bills = df_clean[
    df_clean["Billing Amount"] < 0
]

print("Negative billing amounts:", len(negative_bills))
display(negative_bills[["Billing Amount"]].describe())

# Identify records with missing billing amounts.
missing_bills = df_clean[
```

```
    df_clean["Billing Amount"].isna()
]

print("Missing billing amounts:", len(missing_bills))
display(missing_bills)
```

10.6 Audit Text Formatting

```
# Identify leading or trailing spaces in text columns.
text_columns = df_clean.select_dtypes(
    include="object"
).columns

for col in text_columns:
    spaces = (
        df_clean[col].notna() &
        (df_clean[col] != df_clean[col].str.strip())
    )
    print(col, "- values with surrounding spaces:", spaces.sum())

# Check for possible name variations caused by letter casing.
name_normalised = (
    df_clean["Name"]
    .str.strip()
    .str.casefold()
)

original_distinct_names = df_clean["Name"].nunique()
normalised_distinct_names = name_normalised.nunique()

print("Distinct original names:", original_distinct_names)
print("Distinct normalised names:", normalised_distinct_names)
```

```
print(
    "Difference after normalisation:",
    original_distinct_names - normalised_distinct_names
)

# These results do not prove that records belong to the same person.
```

10.7 Analyse Patient Demographics

```
# Summarise patient ages.
display(df_clean["Age"].describe())

# Count gender records.
print(df_clean["Gender"].value_counts())

# Calculate gender percentages.
print(
    df_clean["Gender"].value_counts(normalize=True) * 100
)

# Count blood types.
print(df_clean["Blood Type"].value_counts())

# Plot the age distribution.
plt.figure(figsize=(8, 5))
sns.histplot(df_clean["Age"], bins=20, kde=True)
plt.title("Distribution of Patient Ages")
plt.xlabel("Age")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()

# Plot gender counts.
```

```
plt.figure(figsize=(6, 4))
sns.countplot(data=df_clean, x="Gender")
plt.title("Patient Records by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()
```

10.8 Analyse Medical Conditions

```
# Calculate medical condition counts and percentages.
condition_counts = df_clean["Medical Condition"].value_counts()

condition_percentages = (
    df_clean["Medical Condition"]
    .value_counts(normalize=True) * 100
)

display(pd.DataFrame({
    "Count": condition_counts,
    "Percentage": condition_percentages
}))

# Plot medical condition frequencies.
plt.figure(figsize=(9, 5))
sns.countplot(
    data=df_clean,
    x="Medical Condition",
    order=condition_counts.index
)
plt.title("Distribution of Medical Conditions")
plt.xlabel("Medical Condition")
plt.ylabel("Number of Records")
```

```
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# Compare billing amounts by medical condition.
condition_billing = (
    df_clean.groupby("Medical Condition")["Billing Amount"]
    .agg(["count", "mean", "median"])
    .round(2)
)

display(condition_billing)

# Plot average billing by condition.
plt.figure(figsize=(9, 5))
sns.barplot(
    data=df_clean,
    x="Medical Condition",
    y="Billing Amount",
    estimator="mean",
    errorbar=None
)
plt.title("Average Billing by Medical Condition")
plt.xlabel("Medical Condition")
plt.ylabel("Average Billing Amount")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```

10.9 Analyse Admission Types

```
# Count admission types.
print(df_clean["Admission Type"].value_counts())
```

```
# Calculate the average length of stay for each admission type.
stay_by_admission = (
    df_clean.groupby("Admission Type")["Length of Stay"]
    .agg(["count", "mean", "median"])
    .round(2)
)

display(stay_by_admission)

# Calculate average billing by admission type.
billing_by_admission = (
    df_clean.groupby("Admission Type")["Billing Amount"]
    .agg(["count", "mean", "median", "std"])
    .round(2)
)

display(billing_by_admission)

# Visualise admission counts.
plt.figure(figsize=(7, 4))
sns.countplot(data=df_clean, x="Admission Type")
plt.title("Number of Records by Admission Type")
plt.xlabel("Admission Type")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()

# Visualise average length of stay.
plt.figure(figsize=(7, 4))
sns.barplot(
```

```
    data=df_clean,
    x="Admission Type",
    y="Length of Stay",
    estimator="mean",
    errorbar=None
)
plt.title("Average Length of Stay by Admission Type")
plt.xlabel("Admission Type")
plt.ylabel("Average Length of Stay (Days)")
plt.tight_layout()
plt.show()
```

10.10 Compare Hospital Groups

```
# Group hospital records and calculate summary statistics.
hospital_summary = (
    df_clean.groupby("Hospital")
    .agg(
        Patient_Count=("Age", "size"),
        Average_Billing=("Billing Amount", "mean"),
        Median_Billing=("Billing Amount", "median"),
        Average_Stay=("Length of Stay", "mean")
    )
    .round(2)
)

# Keep groups with at least 20 records for comparison.
hospital_comparison = hospital_summary[
    hospital_summary["Patient_Count"] >= 20
].copy()

print("Hospital-name groups with at least 20 records:",
      len(hospital_comparison))
```

```
# Display the groups with the highest average billing.
display(
    hospital_comparison
    .sort_values("Average_Billing", ascending=False)
    .head(10)
)

# Display the groups with the lowest average billing.
display(
    hospital_comparison
    .sort_values("Average_Billing", ascending=True)
    .head(10)
)

# Display the groups with the longest average stay.
display(
    hospital_comparison
    .sort_values("Average_Stay", ascending=False)
    .head(10)
)

# Plot average billing for the ten highest-billing groups.
top_hospitals = (
    hospital_comparison
    .sort_values("Average_Billing", ascending=False)
    .head(10)
    .reset_index()
)

plt.figure(figsize=(11, 6))
```

```
sns.barplot(
    data=top_hospitals,
    x="Average_Billing",
    y="Hospital",
    errorbar=None
)
plt.title("Hospital-Name Groups with Highest Average Billing")
plt.xlabel("Average Billing Amount")
plt.ylabel("Hospital Name")
plt.tight_layout()
plt.show()
```

10.11 Analyse Insurance Providers

```
# Calculate billing statistics by insurance provider.
insurance_summary = (
    df_clean.groupby("Insurance Provider")["Billing Amount"]
    .agg(["count", "mean", "median"])
    .round(2)
)

display(insurance_summary.sort_values("mean", ascending=False))

# Plot average billing by insurance provider.
plt.figure(figsize=(9, 5))
sns.barplot(
    data=df_clean,
    x="Insurance Provider",
    y="Billing Amount",
    estimator="mean",
    errorbar=None
)
plt.title("Average Billing by Insurance Provider")
```

```
plt.xlabel("Insurance Provider")
plt.ylabel("Average Billing Amount")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```

10.12 Analyse Medication and Test Results

```
# Count medication records.
print(df_clean["Medication"].value_counts())

# Count test result categories.
print(df_clean["Test Results"].value_counts())

# Calculate test result percentages within each condition.
test_result_percentages = pd.crosstab(
    df_clean["Medical Condition"],
    df_clean["Test Results"],
    normalize="index"
) * 100

display(test_result_percentages.round(2))

# Plot test results by medical condition.
plt.figure(figsize=(10, 6))
sns.countplot(
    data=df_clean,
    x="Medical Condition",
    hue="Test Results"
)
plt.title("Test Results by Medical Condition")
plt.xlabel("Medical Condition")
plt.ylabel("Number of Records")
```

```
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```

10.13 Analyse Billing and Length of Stay

```
# Calculate the Pearson correlation between billing and stay.
correlation = df_clean[
    ["Length of Stay", "Billing Amount"]
].corr()

display(correlation)

# Calculate billing statistics for each length of stay.
billing_by_stay = (
    df_clean.groupby("Length of Stay")["Billing Amount"]
    .agg(["count", "mean", "median"])
    .round(2)
)

display(billing_by_stay)

# Scatter plot to inspect the relationship.
plt.figure(figsize=(9, 5))
sns.scatterplot(
    data=df_clean,
    x="Length of Stay",
    y="Billing Amount",
    alpha=0.4
)
plt.title("Length of Stay Versus Billing Amount")
plt.xlabel("Length of Stay (Days)")
plt.ylabel("Billing Amount")
```

```
plt.tight_layout()
plt.show()

# Line plot of average billing by length of stay.
plt.figure(figsize=(10, 5))
sns.lineplot(
    data=billing_by_stay.reset_index(),
    x="Length of Stay",
    y="mean",
    marker="o"
)
plt.title("Average Billing by Length of Stay")
plt.xlabel("Length of Stay (Days)")
plt.ylabel("Average Billing Amount")
plt.tight_layout()
plt.show()
```

10.14 Final Data Quality Audit

```
# Confirm the final dataset dimensions.
print("Final dataset shape:", df_clean.shape)

# Confirm that duplicate rows have been removed.
print("Remaining duplicates:", df_clean.duplicated().sum())

# Check missing values after preprocessing.
print("\nFinal missing values:")
print(df_clean.isnull().sum())

# Confirm the main validation checks.
print(
    "Invalid ages:",
    ((df_clean["Age"] < 0) | (df_clean["Age"] > 120)).sum()
```

```
)

print(
    "Negative stays:",
    (df_clean["Length of Stay"] < 0).sum()
)

print(
    "Zero-day stays:",
    (df_clean["Length of Stay"] == 0).sum()
)

print(
    "Negative billing amounts:",
    (df_clean["Billing Amount"] < 0).sum()
)
```